Final Exam

Due: 7:30pm on Wednesday, June 30, 2021

You will receive no credit for items you complete after the assignment is due. Grading Policy

Q 1

Consider a circular coil. It has a diameter of 3.80 cm and has 540 turns. The circular coil carries a current of 0.560 A .

Part A

Calculate the magnitude of the magnetic field at the center of the circular coil?

Express your answer in teslas.

ANSWER:

B = 1.00×10^{-2} T

Incorrect; correct answer displayed

Part B

Find the magnitude of the magnetic field at a point on the central axis of the coil a distance of 6.80 cm from its center?

Express your answer in teslas.

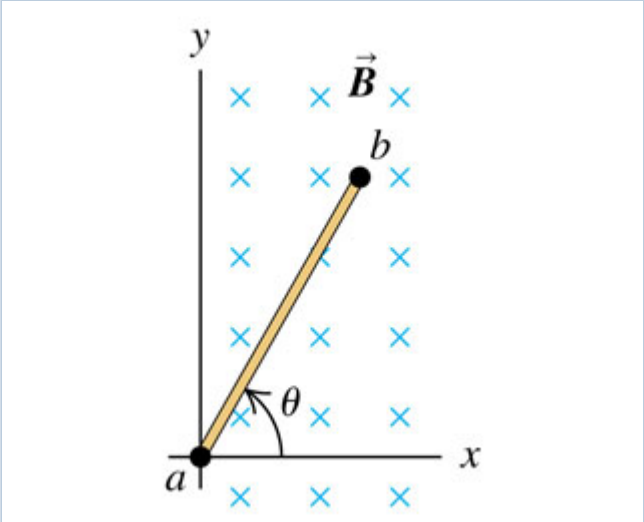
ANSWER:

B = 1.95×10^{-4} T

Incorrect; correct answer displayed

Q 2

A uniform bar has mass 0.0160 kg and is 30.0 cm long. It pivots without friction about an axis perpendicular to the bar at point a in . The gravitational force on the bar acts in the $-y$-direction. The bar is in a uniform magnetic field that is directed into the page and has magnitude 0.300 B .



Part A

Calculate the torque on the bar due to gravitation when it is at an angle 45.0 ° above the horizontal?

ANSWER:

1.66×10^{-2}

Incorrect; correct answer displayed

Part B

What must be the current I in the bar for the bar to be in rotational equilibrium when it is at an angle 45.0 ° above the horizontal?

Express your answer with the appropriate units.

ANSWER:

I = 1.23 A

Incorrect; correct answer displayed

Part C

For the bar to be in rotational equilibrium, should I be in the direction from *a* to *b* or *b* to *a*?

Drag the terms on the left to the appropriate blanks on the right to complete the sentences.

ANSWER:

Reset Help

counterclockwise

a

upward

clockwise

the right

b

downward

the left

Gravity tends to rotate the bar **clockwise** about point *a*, so the magnetic force must be **upward** and to **the left** to tend to rotate the bar **counterclockwise**. Therefore the current must flow from a to b. If the current were from b to a, the bar could not balance.

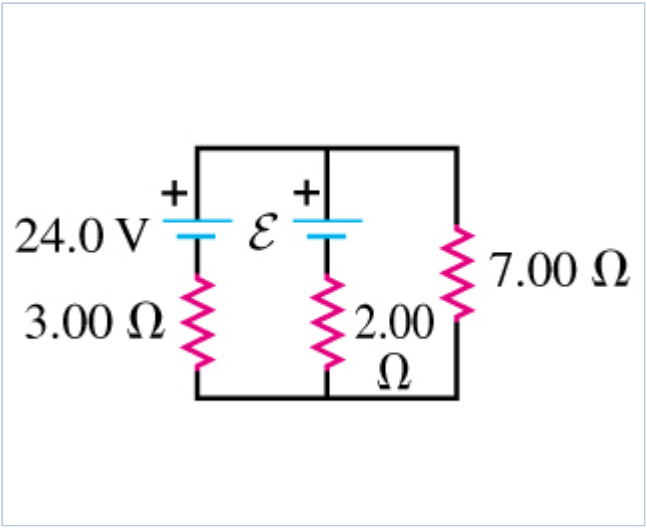
Correct

Q 3

Part A

Find the emf $\mathcal{E}$ in the figure so that the current through the 7.00 Ω resistor must be 1.85 A ? Assume no internal resistance for all emf sources.

Express your answer with the appropriate units.



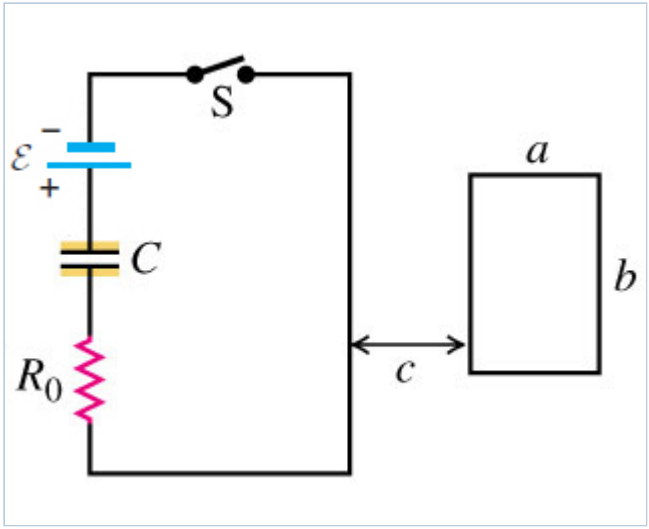
ANSWER:

$\mathcal{E}$ = 9.28 V

Incorrect; correct answer displayed

Q 4

In the circuit shown in the following figure the capacitor has capacitance 23 $\mu\mathrm{F}$ and is initially uncharged. The resistor R_0 has resistance 10 Ω . An emf of 92.0 V is added in series with the capacitor and the resistor. The emf is placed between the capacitor and the switch, with the positive terminal of the emf adjacent to the capacitor. The small circuit is not connected in any way to the large one. The wire of the small circuit has a resistance of 1.5 Ω/m and contains 20 loops. The large circuit is a rectangle 2.0 m by 4.0 m, while the small one has dimensions a = 12.0 cm and b = 25.0 cm . The distance c is 3.0 cm . (The figure is not drawn to scale.) Both circuits are held stationary. Assume that only the wire nearest the small circuit produces an appreciable magnetic field through it.



Part A

The switch is closed at t = 0. When the current in the large circuit is 3.50 A , what is the magnitude of the induced current in the small circuit?

Express your answer with the appropriate units.

ANSWER:

I = 1.1×10^{-3} A

Incorrect; correct answer displayed

Part B

What is the direction of the induced current in the small circuit?

ANSWER:

- (•) clockwise
- ( ) counterclockwise

Correct

Q 5

A coil with magnetic moment 1.50 $\mathrm{A}\cdot\mathrm{m}^2$ is oriented initially with its magnetic moment antiparallel to a uniform magnetic field of magnitude 0.825 T .

Part A

What is the change in potential energy of the coil when it is rotated 180 degrees, so that its magnetic moment is parallel to the field?

Express your answer in joules.

ANSWER:

-2.48 J

Incorrect; correct answer displayed

Q 6

A point charge of 25.0 nC is placed in a uniform electric field that is directed vertically upward and has a magnitude of $3.90\times10^4\ \mathrm{V/m}$.

Part A

Find the work done by the electric force when the charge moves a distance of 0.500 m to the right?

Express your answer in joules.

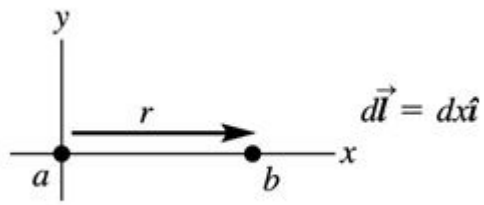
ANSWER:

W = 0 J

Incorrect; correct answer displayed

IDENTIFY: Apply $W_{a\to b} = q'\int_a^b \vec{E}\cdot d\vec{l}$. Use coordinates where $+y$ is upward and $+x$ is to the right. Then $\vec{E} = E\hat{j}$ with $E = 3.90\times10^4\ \mathrm{V/m}$.

SET UP: (a) The path is sketched in the figure below.



EXECUTE: $\vec{E}\cdot d\vec{l} = (E\hat{j})\cdot(dx\hat{i}) = 0$ so $W_{a\to b} = q'\int_a^b \vec{E}\cdot d\vec{l} = 0\ \mathrm{J}$

EVALUATE: The electric force on the positive charge is upward (in the direction of the electric field) and does no work for a horizontal displacement of the charge.

Part B

Find the work done by the electric force when the charge moves a distance of 0.640 m upward?

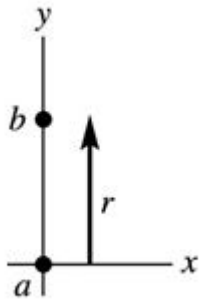
Express your answer in joules.

ANSWER:

W = 6.24×10^{-4} J

Incorrect; correct answer displayed

SET UP: The path is sketched in the figure below.



$d\vec{l} = dy\hat{j}.$

EXECUTE: $\vec{E}\cdot d\vec{l} = (E\hat{j})\cdot(dy\hat{j}) = Edy$

$$W_{a\to b} = q'\int_a^b \vec{E}\cdot d\vec{l} = q'E\int_a^b dy = q'E(y_b - y_a)$$

$y_b - y_a = 0.640\ \mathrm{m}$; it is positive since the displacement is upward and we have taken $+y$ to be upward.

$$W_{a\to b} = q'E(y_b - y_a) = (+25.0\times10^{-9}\ \mathrm{C})(3.90\times10^{4}\ \mathrm{V/m})(0.640\ \mathrm{m}) = 6.24\times10^{-4}\ \mathrm{J}$$

EVALUATE: The electric force on the positive charge is upward so it does positive work for an upward displacement of the charge.

Part C

Find the work done by the electric force when the charge moves a distance of 2.80 m at an angle of 45.0° downward from the horizontal?

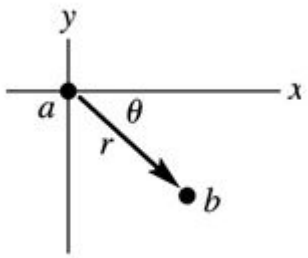
Express your answer in joules.

ANSWER:

W = -1.93×10^{-3} J

Incorrect; correct answer displayed

SET UP: The path is sketched in the figure below.



$y_a = 0$

$y_b = -r\sin\theta = -(2.80\ \text{m})\sin 45.0^\circ = -1.980\ \text{m}$

The vertical component of the $2.80\ \text{m}$ displacement is $1.980\ \text{m}$ downward.

EXECUTE: $d\vec{l} = dx\hat{i} + dy\hat{j}$ (The displacement has both horizontal and vertical components.)

$\vec{E}\cdot d\vec{l} = (E\hat{j})\cdot(dx\hat{i} + dy\hat{j}) = Edy$ (Only the vertical component of the displacement contributes to the work.)

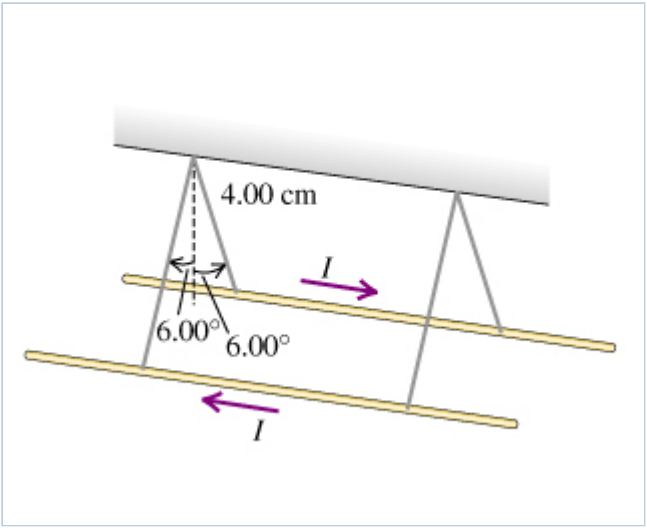
$$W_{a\to b} = q'\int_a^b \vec{E}\cdot d\vec{l} = q'E\int_a^b dy = q'E(y_b - y_a)$$

$$W_{a\to b} = q'E(y_b - y_a) = (+25.0\times 10^{-9}\ \text{C})(3.90\times 10^4\ \text{V/m})(-1.980\ \text{m}) = -1.93\times 10^{-3}\ \text{J}$$

EVALUATE: The electric force on the positive charge is upward so it does negative work for a displacement of the charge that has a downward component.

Q 7

Parallel wires hang by 4.00-cm-long cords from a common axis (see the figure). The parallel wires are too long. Both wires have a mass per unit length of $1.75\times10^{-2}\ \text{kg/m}$. They carry the same current but in opposite directions.



Part A

Find the current in each wire if the cords hang at an angle of 6.00 $^\circ$ with the vertical?

Express your answer in amperes.

ANSWER:

I = 27.5 A

Incorrect; correct answer displayed

Q 8

A dielectric-filled parallel-plate capacitor has plate area A = 30.0 cm^2 , plate separation d = 10.0 mm and dielectric constant k = 3.00. The capacitor is connected to a power supply that creates a constant voltage V = 5.00 V . Throughout the problem, use ϵ_0 = 8.85×10^{-12} $\mathrm{C}^2/\mathrm{N}\cdot\mathrm{m}^2$.

Part A

What is the energy U_1 of the dielectric-filled capacitor.

Express your answer numerically in joules.

ANSWER:

U_1 = 9.96×10^{-11} J

Incorrect; correct answer displayed

Part B

The dielectric slab is now slowly pulled out of the capacitor, which remains connected to the battery. What is the energy U_2 of the capacitor at the moment when the capacitor is half-filled with the dielectric.

Express your answer numerically in joules.

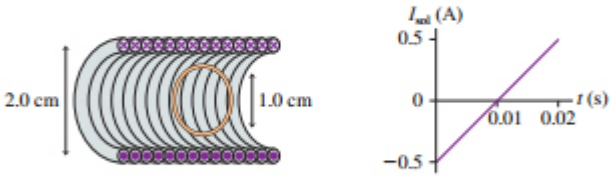
ANSWER:

U_2 = 6.64×10^{-11} J

Incorrect; correct answer displayed

Q 9

shows a 1.0-cm-diameter loop with R = 0.50 Ω inside a 2.0-cm-diameter solenoid. The solenoid is 8.0 cm long, has 130 turns, and carries the current shown in the graph. A positive current is cw when seen from the left.



Part A

Determine the current in the loop at t = 0.010 s?

Express your answer with the appropriate units.

ANSWER:

I = 1.6×10^{-5} A

Incorrect; correct answer displayed

Q 10

Consider a copper wire with a lenght of 3.00 m at 20° C. It has a 1.20-m-long section with diameter 1.80 mm and a 1.80-m-long section with diameter 0.90 mm. There is a current of 2.2 mA in the 1.80-mm-diameter section. (for coper: ρ=1.72x10^{-8} Ω.m)

Part A

Find the current in the 0.90-mm-diameter section?

Express your answer in milliamperes.

ANSWER:

I_2 = 2.2 mA

Incorrect; correct answer displayed

Part B

Calculate the potential difference between the ends of the 3.00 m length of wire?

Express your answer in volts.

ANSWER:

$V = 1.2\times10^{-4}\ \text{V}$

Incorrect; correct answer displayed

Score Summary:

Your score on this assignment is 8.3%.
You received 0.83 out of a possible total of 10 points.